

JEE MAIN REVISION MASTER: WORK, ENERGY AND POWER

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Work-Energy Theorem & Conservative Forces	7 PYQs	Work & Energy
Moderate	Power & Collisions (Elastic and Inelastic)	3 PYQs	Collisions & Power

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Work-Energy Theorem & Conservative Forces	$W = \int F \, dx = \Delta K$, $U_{\text{spring}} = \frac{1}{2} k x^2$, $P = \vec{F} \cdot \vec{v} = \frac{dW}{dt}$
Power & Collisions (Elastic and Inelastic)	$e = \frac{v_2 - v_1}{u_1 - u_2}$, $v_{\text{common}} = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$, $\Delta K = \frac{m_1 m_2 (1 - e^2)}{2(m_1 + m_2)}(u_1 - u_2)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Work-Energy Theorem & Conservative Forces (Moderate Yield — 7 PYQs)

Core Principle: Work-Energy Theorem: $W_{\text{net}} = \Delta K = K_f - K_i$. Work done by variable force is area under F-x curve. Potential energy of spring $U = \frac{1}{2} k x^2$. Mechanical energy $E = K + U$ is conserved under conservative forces.

■ **Caution / JEE Pitfall:** Work done by a perpendicular force (like centripetal force or magnetic Lorentz force) is strictly ZERO ($W=0$).

■ **Speed Trick:** Relation between kinetic energy K and momentum p : $K = (p^2 / 2m)$. If momentum increases by 100%, kinetic energy increases by 300%!

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A horizontal force of 5 N is applied on a stationary body of mass 5 kg, which is initially at rest on a frictionless table. The change i..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Two bodies with kinetic energies in the ratio of 3:1 are moving with equal linear momentum. The ratio of their masses is..." [Ans: B]
- [2025 Shift] Q#3 (MCQ): "A body of mass 0.25 kg travels along a straight line from $x=0$ to $x=2m$ with a speed $v=kx^{3/2}$ where $k=2SI$ units. The work done by..." [Ans: N/A]

3.2 Power & Collisions (Elastic and Inelastic) (Moderate Yield — 3 PYQs)

Core Principle: Instantaneous power $P = \text{vec}(F) \cdot \text{vec}(v)$. Coefficient of restitution $e = (\text{velocity of separation} / \text{velocity of approach})$. In 1D elastic collision ($e=1$) between equal masses ($m_1=m_2$), velocities are completely exchanged! In perfectly inelastic collision ($e=0$), bodies stick together with common speed $v = (m_1 u_1 + m_2 u_2 / m_1 + m_2)$.

■ **Caution / JEE Pitfall:** Momentum is conserved in ALL collisions (elastic or inelastic); kinetic energy is conserved ONLY in perfectly elastic collisions.

■ **Speed Trick:** When a moving body hits an identical stationary body elastically ($m_1=m_2, u_2=0$), the incident body stops dead ($v_1=0$) and the target moves off with speed $v_2 = u_1$.

Representative Source PYQs:

- **[2023 Shift] Q#1 (MCQ):** "A ball of mass 0.2kg is thrown vertically down from a height of 10m . It collides with the floor and loses 50% of its energy and then ..." **[Ans: N/A]**
- **[2020 Shift] Q#3 (MCQ):** "A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time t is propor..." **[Ans: N/A]**
- **[2017 Shift] Q#1 (MCQ):** "A motor pump lifts 6 tonnes of water from a well of depth 25 m to the first floor of height 35 m from the ground floor in 20 minutes. The po..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A horizontal force of 5 N is applied on a stationary body of mass 5 kg, which is initially at rest on a frictionless table. The change in kinetic energy of the body in 10 s is

- (A) 25 J
- (B) Zero
- (C) 125 J
- (D) 250 J

Correct Answer: None

Solution Analysis: The body is on a frictionless table, so the applied force produces acceleration without any loss due to friction. Given, $F=5\text{N}$ $m=5\text{kg}$ Using Newton's second law, $F=ma$ So, $a=(F / m)=(5 / 5)=1\text{m s}^{-2}$ The body starts from rest, so $u=0$ Time, $t=10\text{s}$ Using $v=u+at$ we get $v=0+1 \times 10=10\text{m s}^{-1}$ Initial kinetic energy is zero because the body is initially at rest. Final kinetic energy is $K=(1 / 2)mv^2 K=(1 / 2) \times 5 \times (10)^2 K=(1 / 2) \times 5 \times 100=250\text{J}$ So, the change in kinetic energy of the body is

Benchmark Problem 2: KCET 2023 (Q#1)

A ball of mass 0.2kg is thrown vertically down from a height of 10m . It collides with the floor and loses 50% of its energy and then rises back to the same height. The value of its initial velocity is

- (A) Zero
- (B) 14ms^{-1}
- (C) 196ms^{-1}
- (D) 20ms^{-1}

Correct Answer: None

Solution Analysis: Given, mass of the ball, $m=0.2\text{kg}$ Height from the surface of floor, $h=10\text{m}$ Let initial velocity be u .
Total energy as initial point $TE=\frac{1}{2}mv^2+mgh$ After collision, remaining energy $=\frac{1}{2}mu^2+mgh$ With this remaining energy the ball bounces upto height h Therefore, $\frac{1}{2}mu^2+mgh=mgh \Rightarrow \frac{1}{2}mu^2+mgh=2mgh \Rightarrow u^2=\sqrt{2gh}=\sqrt{2 \times 10 \times 10}=14\text{m/s}$